SCOPE - X

Electronics at Home

User Manual

Complete Laboratory Manual

• Bipolar Supply ± 12V added • Offset and Gain adjust simplified • Additional breadboard space

TABLE OF CONTENTS

Contents

1 Introduction

Overview: The Scope-X development board from Bumblebee Instruments provides a comprehensive platform for understanding basic analog circuit design principles through hands-on experimentation and analysis.

This manual serves as both an instructional guide and reference document for the Scope-X. Designed by Bumblebee Instruments, the kit provides students and engineers with practical experience in analog circuit design, analysis, and implementation using industry-standard components and methodologies.

The laboratory exercises contained within this manual progress from fundamental operational amplifier concepts to advanced signal processing techniques, ensuring a thorough understanding of analog system design principles.

1.1 Learning Objectives

Upon completion of this manual, users will be able to:

- Design and analyze operational amplifier circuits
- Implement active filter topologies
- Understand analog signal processing fundamental
- Troubleshoot analog circuit implementations

1.2 Kit Components

Table 1: Hardware Components Overview

Component	Description	Quantity
Scope-X Main Board	Analog signal acquisition and processing unit	1
Power Cable	Supplies regulated power to the main board	1
USB Interface Cable	Enables communication with the PC	1

2 System Setup and Configuration

Important: Ensure all power connections are verified before energizing the system. Observe proper ESD handling procedures when working with electronic components.

2.1 Initial Setup Procedure

1. Carefully unpack all kit components and verify contents against the packing list
2. Install the Scope-X software package on your computer
3. Connect the main board to your computer using the provided USB cable
4. Connect the power cable to the board and verify LED indicators

System Requirements

- Operating System: Windows 10/11

2.1.1 Install Application and Drivers

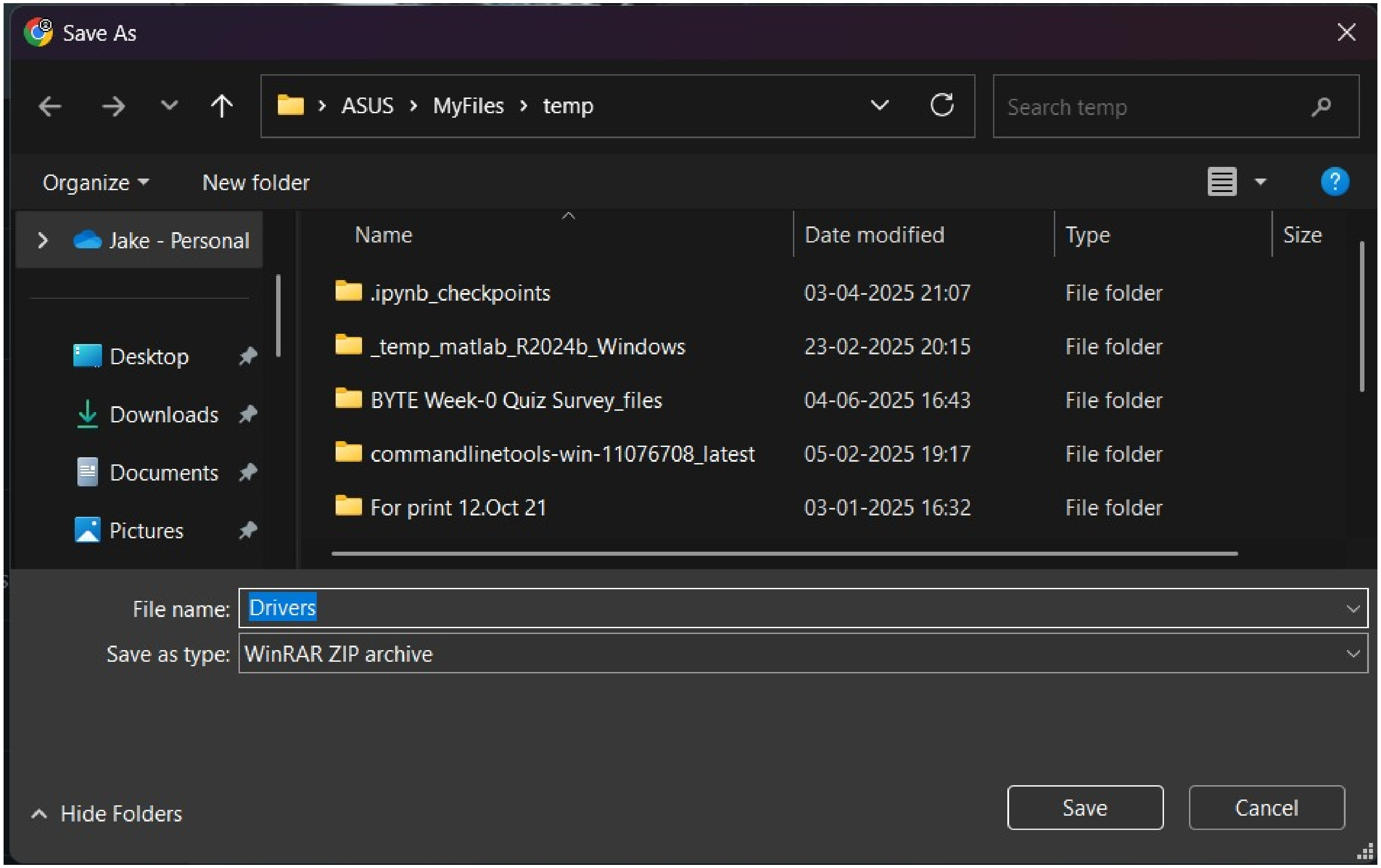

Figure 1: Install the .exe and the Drivers.zip file in your preferred directory.

2.1.2 Open Executable

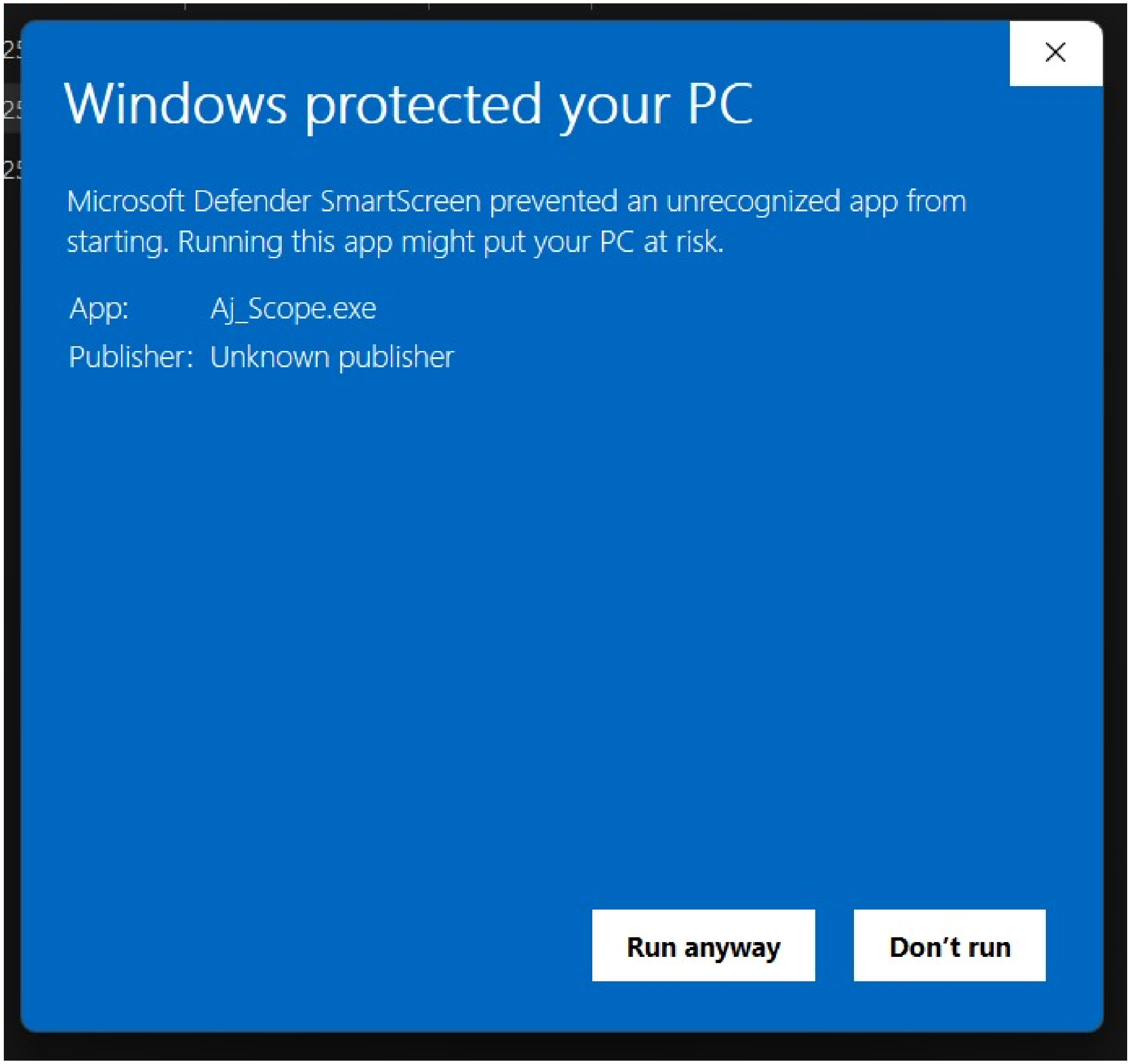

Figure 2: Open the .exe file and select 'Run anyway' if prompted.

2.1.3 Connect Scopex and Find Device

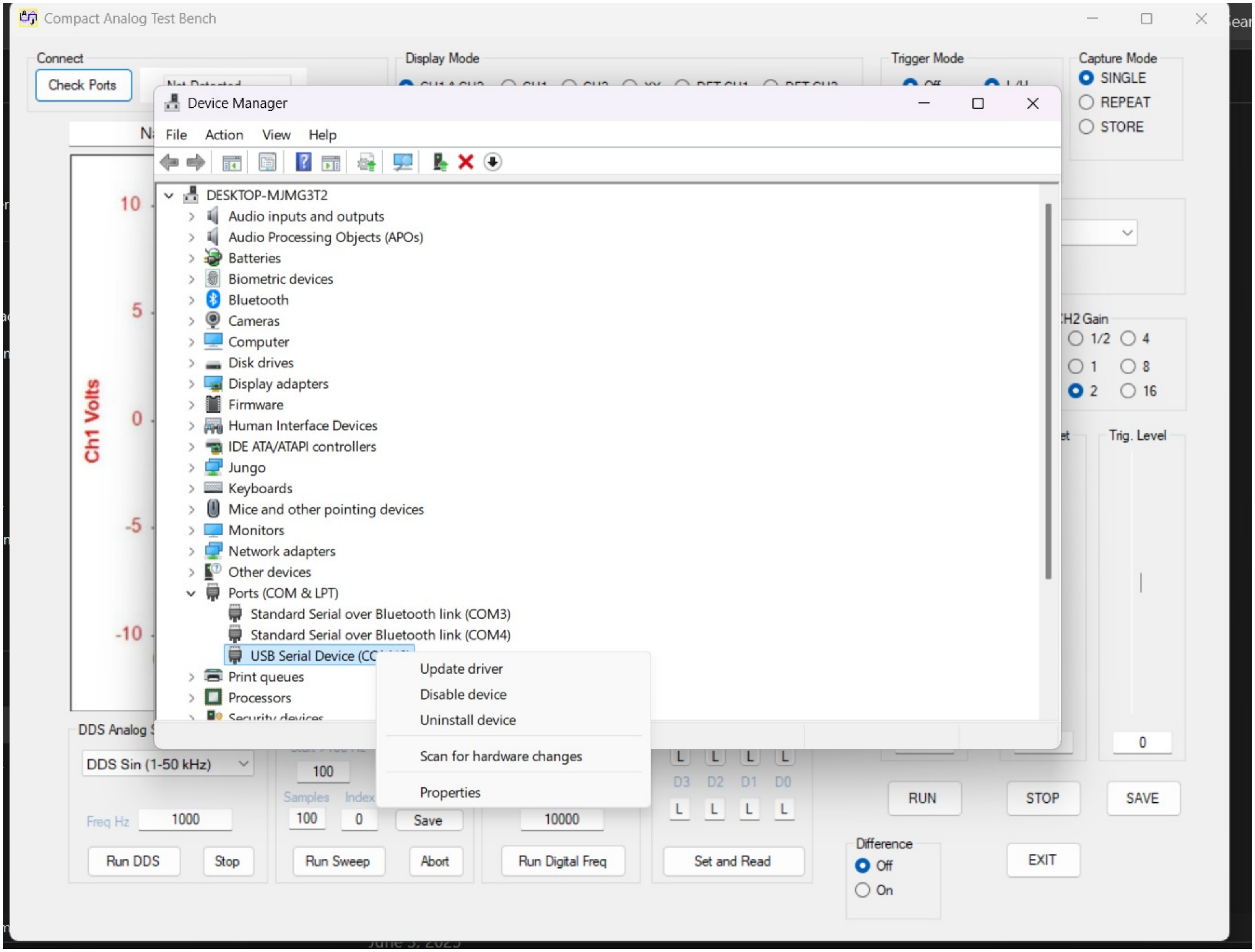

Figure 3: Connect Scopex and open Device Manager. Locate 'USB Serial Device' under COM Ports.

2.1.4 Update Driver

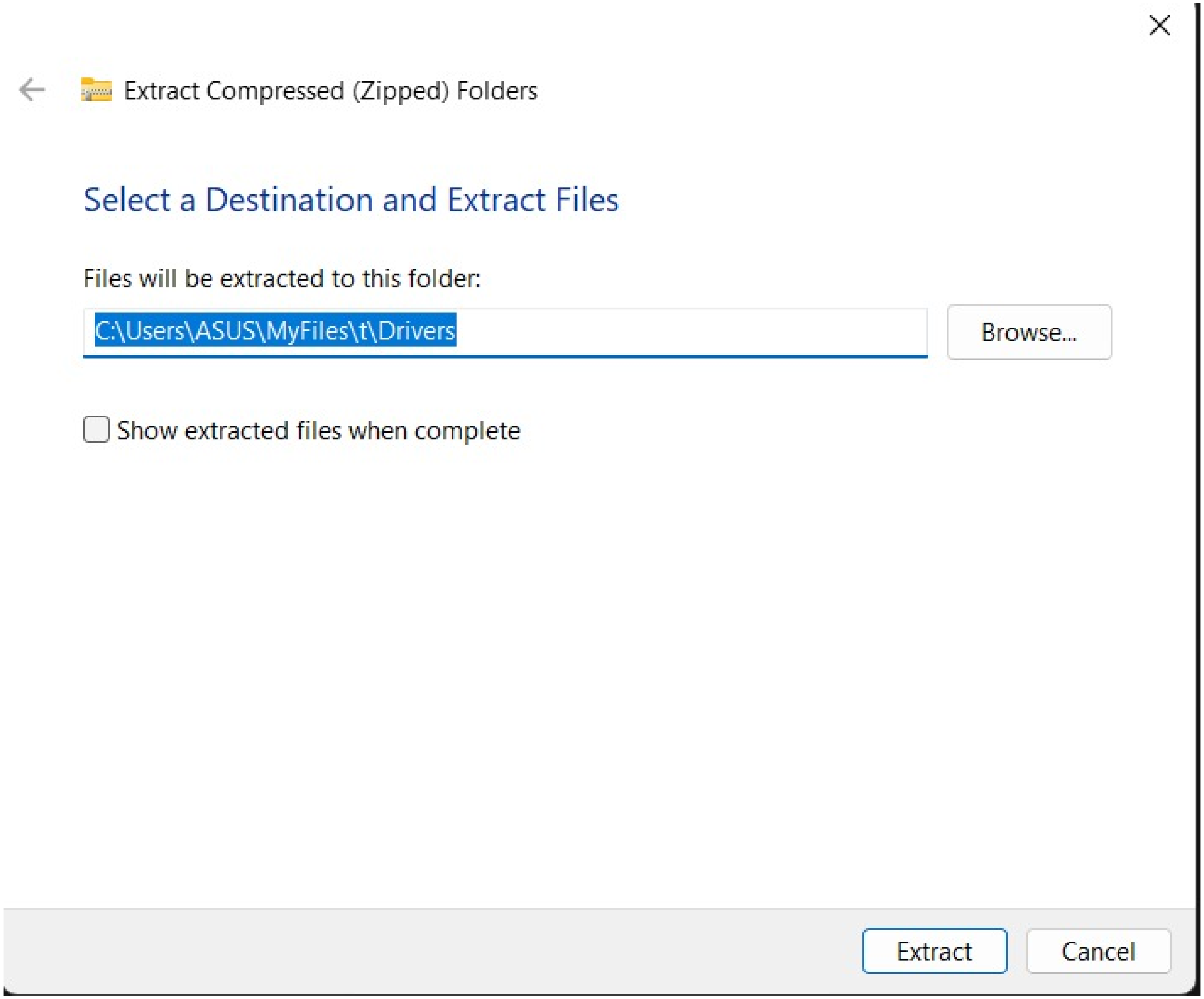

Figure 4: Right-click 'USB Serial Device', select 'Update Driver', and choose the driver folder.

2.1.5 Install Driver

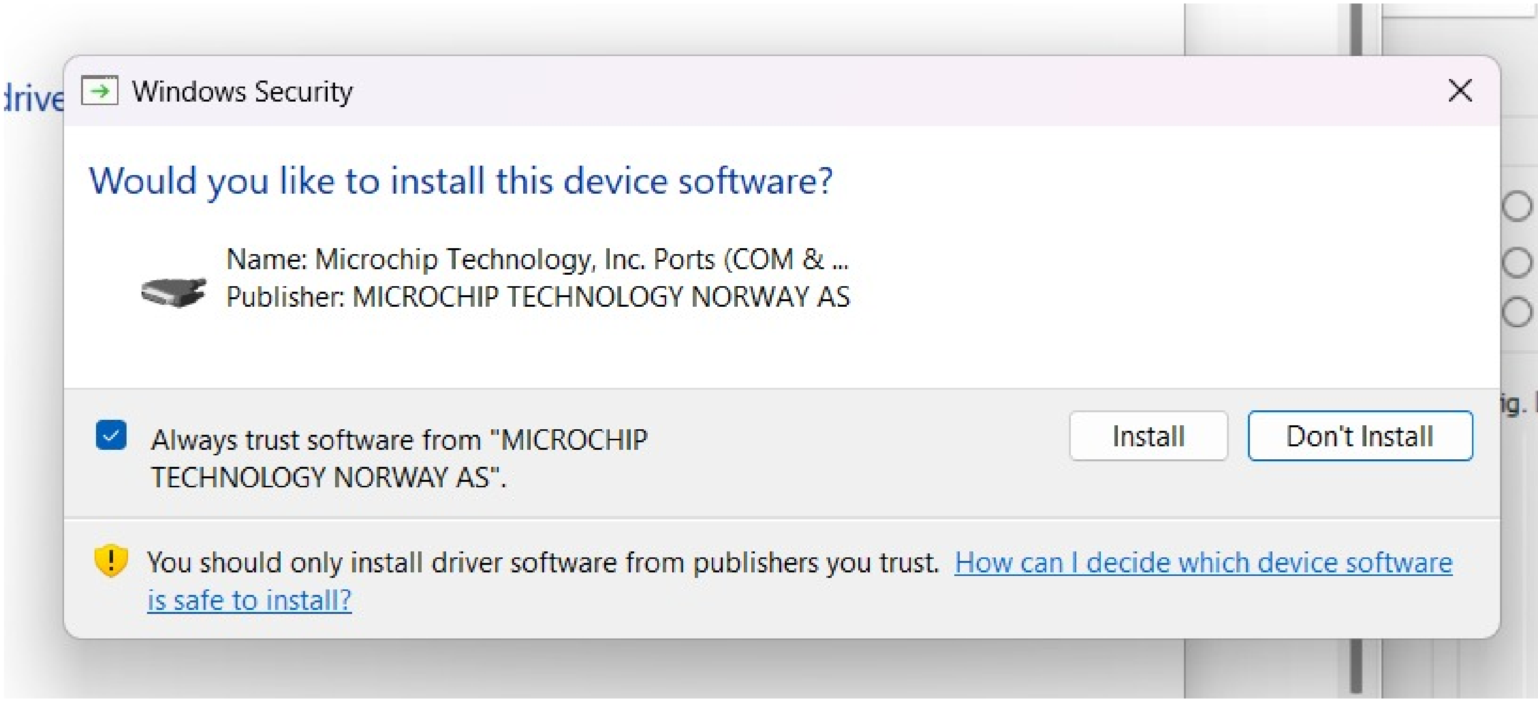

Figure 5: Install the drivers as shown.

2.1.6 Verify COM Port Recognition

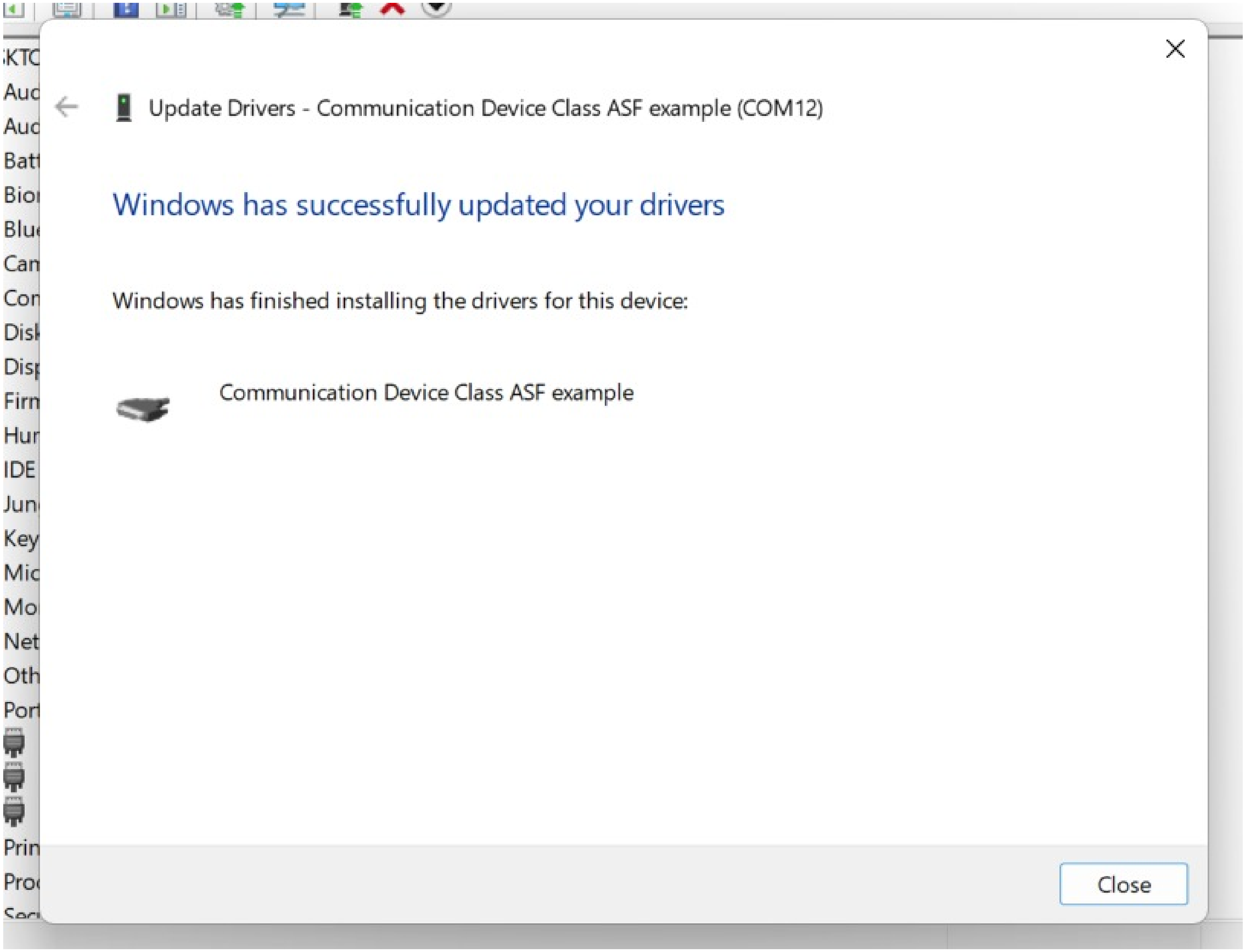

Figure 6: Once installed, verify that Scopex is recognized under the correct COM Port.

2.1.7 Launch GUI and Check Ports

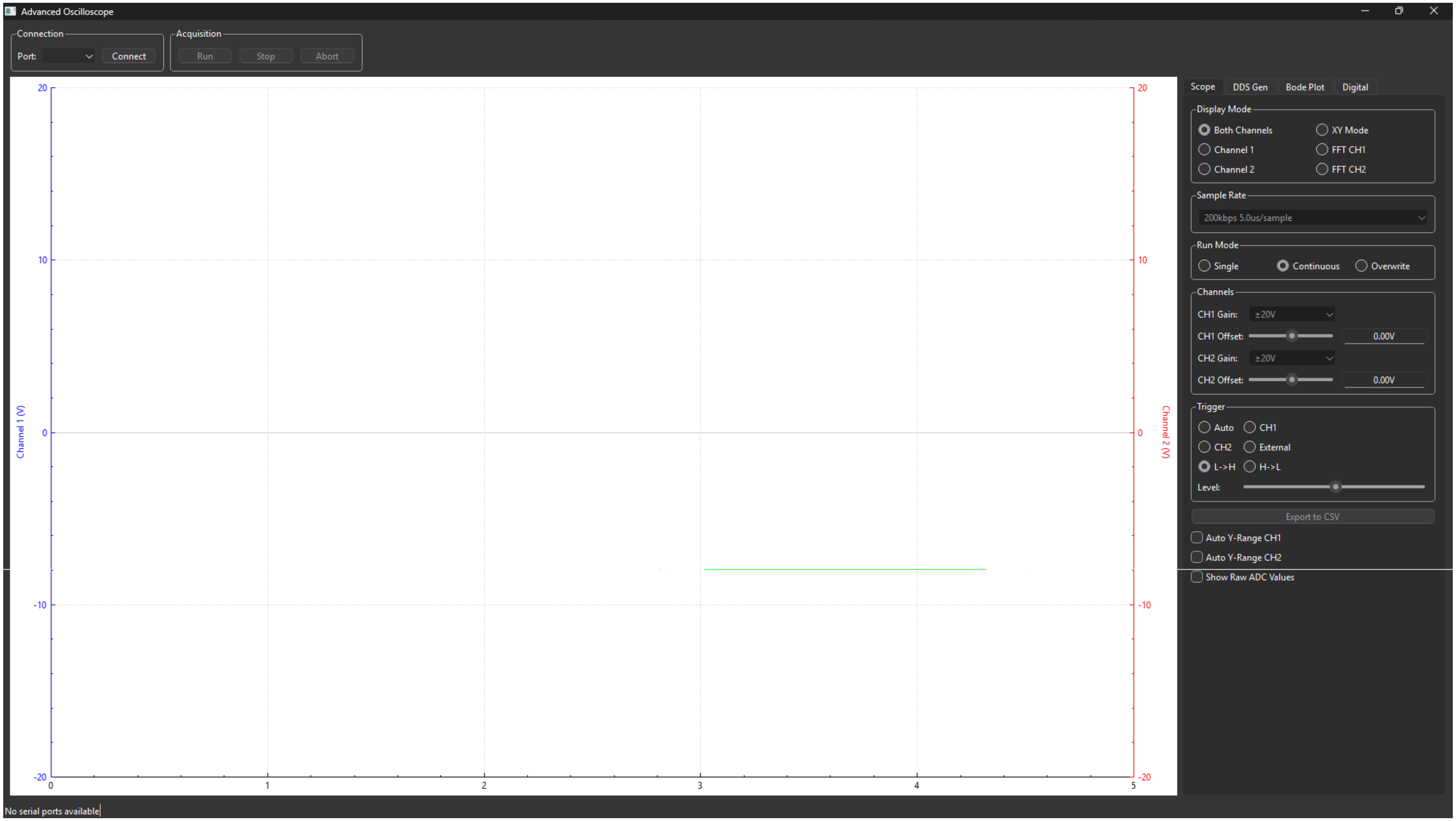

Figure 7: Launch the GUI and click 'connect'.The device is now successfully connected.

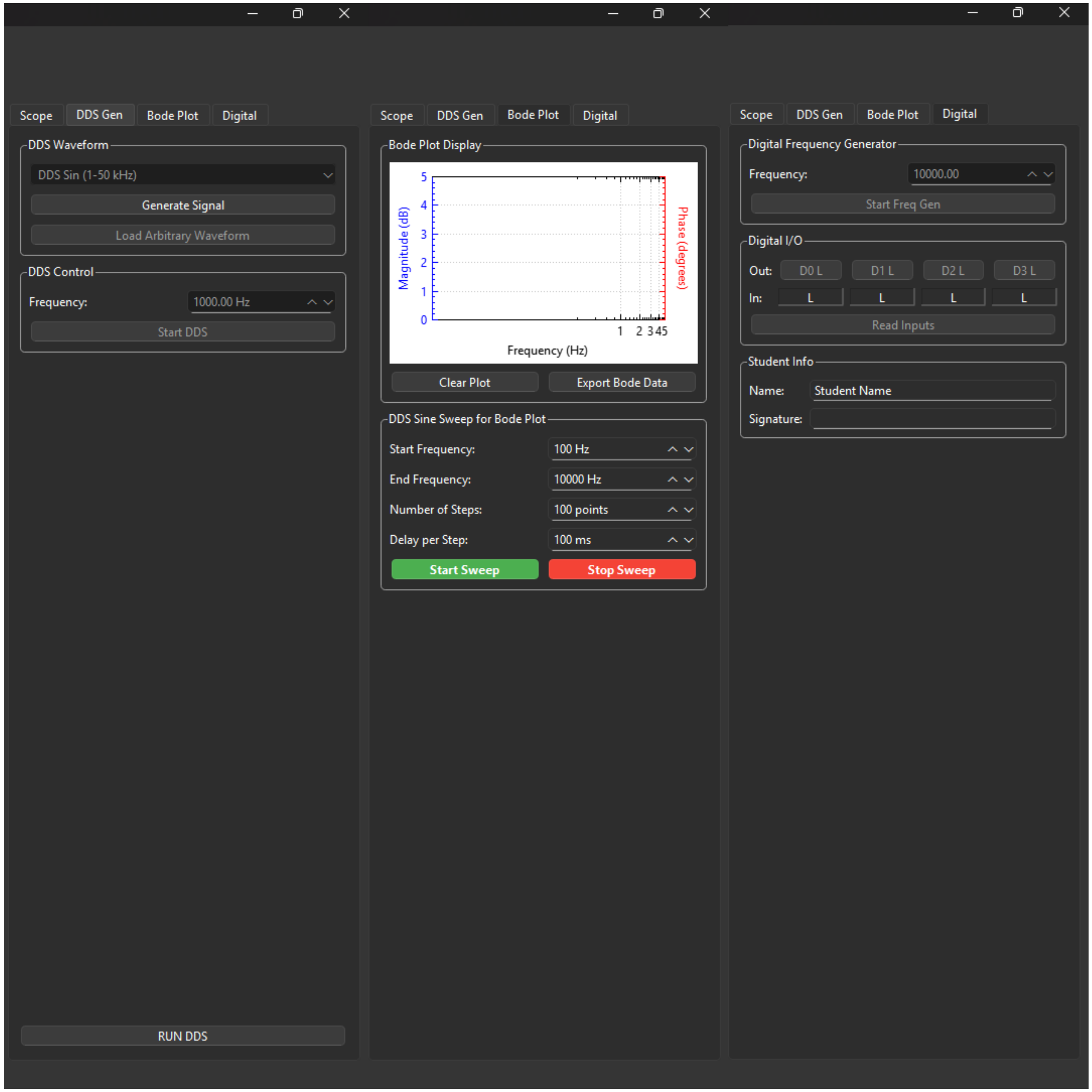

Figure 8: Other taps

3 Hardware Architecture

3.1 Main Board Overview

The Scope-X main board features a modular architecture optimized for educational use and rapid prototyping.

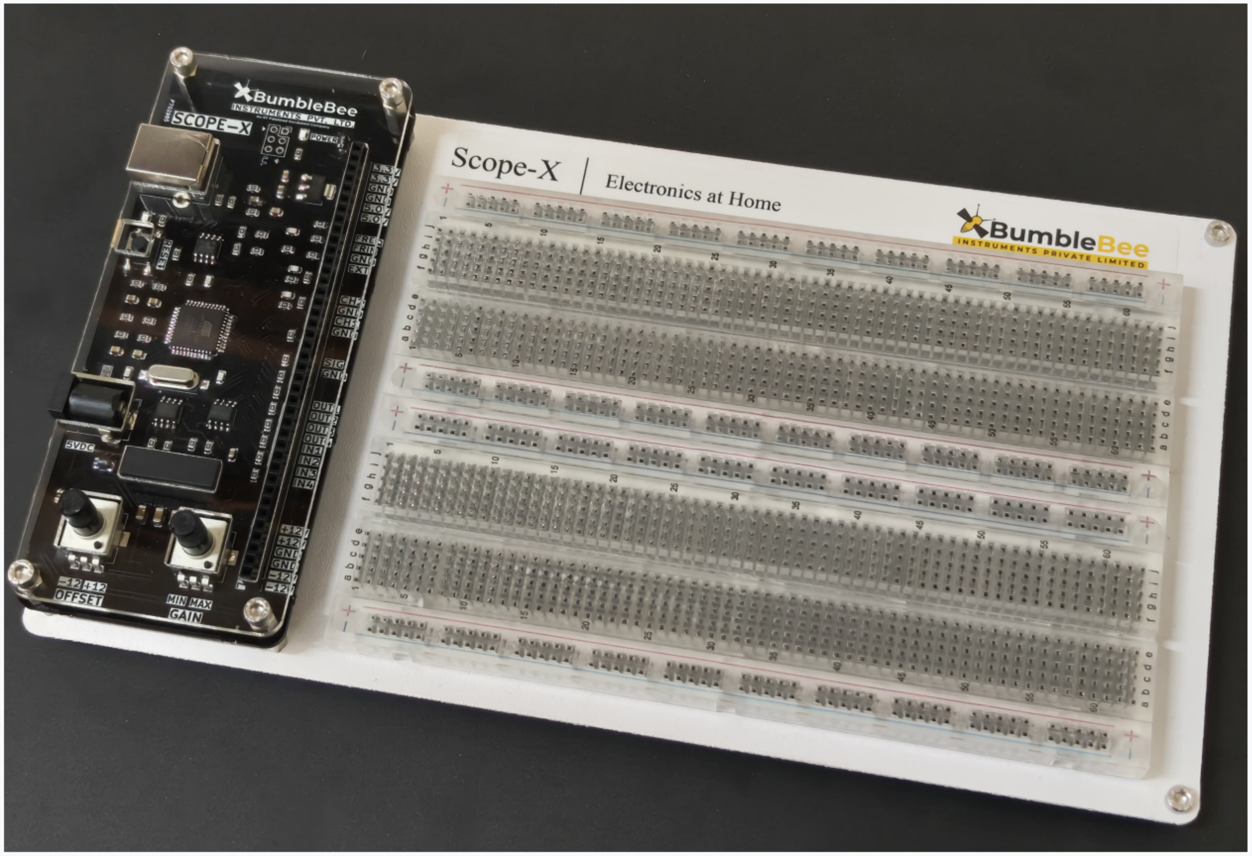

Figure 9: Scope-X Main Board

3.2 Board Features

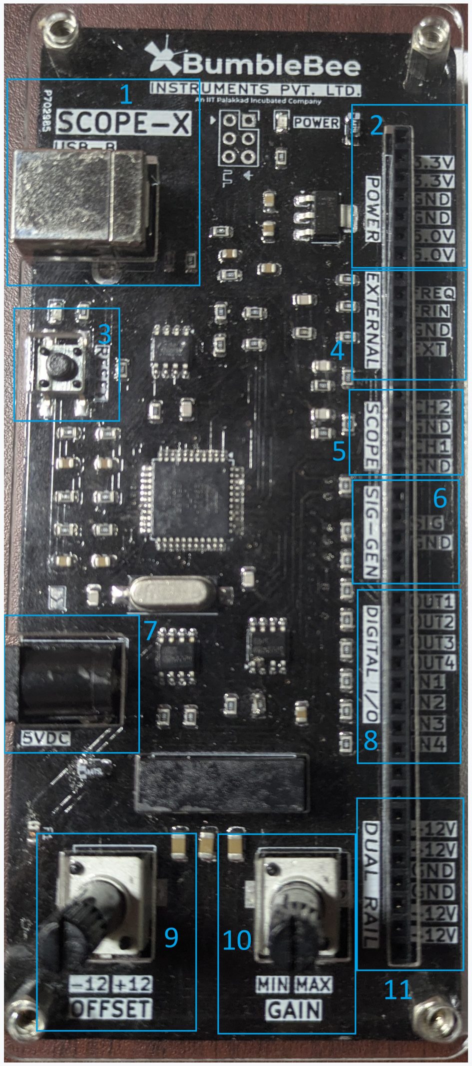

Figure 10: Scope-X Main Board

Table 2: Functions

Features	Usage
1 USB-B	USB adapter to connect with the PC
2 Power	Acts as power supply
3 Reset	Resets the Board
4 External	Other External pins
5 Scope	Oscilloscope with two channels
6 Signal generator	Generates signal that is selected in the GUI
7 5VDC	Power supply to the Board
8 Digital I/O	4 input and output pins to read/generate digital signals
9 Offset	Adjustable Offset between -12V and 12 V
10 Min Max Gain	Adjusts Gain
11 Dual Rail	Bipolar Power supply

4 GUI Software functions

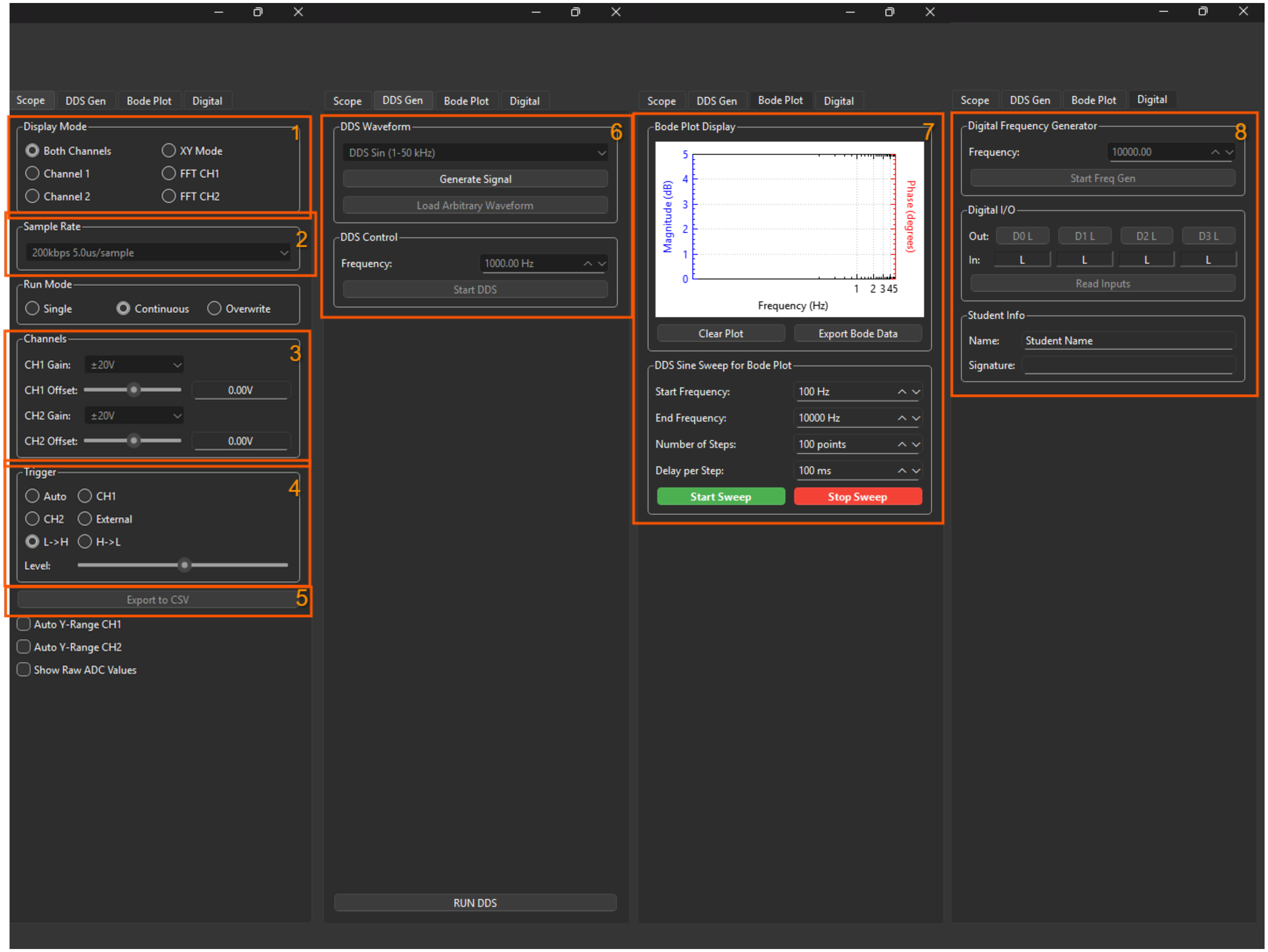

Figure 11: Scope-X GUI

Index	Function
1	**Display Mode** - For the Oscilloscope Mode, X-Y Mode or DFT Mode selects channel to be displayed (CH1 & CH2, CH1, CH2, XY, DFT CH1, DFT CH2)
2	**Sampling Rate** - Selects Sampling rate / Time per sample (200 samples are displayed). Display time scale will change appropriately (200kbps 5.0us/sample shown)
3	**CH1,CH2 Scale** - Sets the Gain for CH1,CH2 (1/2, 1, 2 with corresponding scale values 4, 8, 16). Display amplitude scale will change appropriately **CH1,CH2 Offset** - CH1,CH2 Offset (useful for centering the display around Zero to display small signals with large offset at higher gain values)
4	**Trigger Mode** - Selects the Oscilloscope Trigger mode (Off, CH1, CH2, EXT) and Trigger slope (L/H (Low to High), H/L (High to Low)) **Trig. Level** - Set the trigger threshold in CH1/CH2 trigger modes. (If not triggered display will show Busy)
5	**Capture** - Single – Single shot mode (useful for capturing single events with trigger). Repeat – Continuous running with repeated display. Store – Useful for capturing additional traces (run single then store)
6	**DDS Analog Signal Generator** - DDS-Analog signal generator. Select frequency (1000 Hz shown) and waveform type (DDS Sin 1-50 Hz) and Run. For Arbitrary waves read a single 256 column .csv file with data value 5-250
7	**Bode Plot Generator** -Generates a frequency sweep and plots the system's magnitude and phase response versus frequency for gain and stability analysis.
8	**Digital Freq Generator** - Enter any frequency value 1-100k Hz (10000 shown) and Run to get a 3.3V square-wave / inverted square wave at the digital frequency out pins

5 Troubleshooting and Maintenance

5.1 Common Issues and Solutions

Table 4: Troubleshooting Guide

Problem	Solution
No power indication on main board	Verify power cable connections
Board not recognized	Make sure the drivers are installed and the Scope-X board is recognized as Communication Class Device ASF Example in Device Manager
GUI malfunctioning	In case of any malfunctioning, please restart the application.